



Exam Code: PQ1ABC



UNIVERSITY OF COLOMBO, SRI LANKA

UNIVERSITY OF COLOMBO SCHOOL OF COMPUTING

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination – Semester I – 2017

SCS 1104 – Mathematical Methods I

TWO (2) HOURS

To be completed by the candidate

Examination Index No: _____

Important Instructions to candidates:

1. The medium of instruction and questions is **English**.
2. Note that questions appear on both sides of the paper. If a page or a part of the question paper is not printed, please inform the supervisor immediately.
3. Write your index number on each and every page of the question paper.
4. This paper has **04** questions in **7** pages.
5. Answer **ALL** questions.
6. MCQ TYPE: **Copy the exam code clearly on to your MCQ answer sheet.** For each of the multiple choice questions, encircle the correct response clearly with a pen on the answer sheet that is provided.
7. ESSAY TYPE: Write the answers to all questions on the writing paper that is provided.
8. Any electronic device capable of storing and retrieving text including electronic dictionaries and mobile phones are **not allowed**.
9. Calculators are **not allowed**.

For Examiner's use only

Question No	Marks
1	
2	
3	
4	
Total	

01. Answer the following multiple choice questions on the answer sheet provided.

1. Find the set of solutions to the inequality $\sqrt{(x-3)^2} < 3$.

- (a) $(-3, 0) \cup (0, 3)$
- (b) $(0, 6)$
- (c) $(-3, 3)$
- (d) $(-6, 0)$
- (e) None of the above.

2. Which of the following statements is/are true about the set $A = [3, 6)$?

- (I) The maximum of A is 6.
- (II) The supremum of A is 6, but it is not the maximum.
- (III) minimum and the infimum of A is 3.

- (a) I only
- (b) II only
- (c) I and III only
- (d) II and III only
- (e) None of the above

3. Which of the following statements regarding the function $f(x) = \frac{x}{|x|}$ is true?

- (a) $\lim_{x \rightarrow 0^+} f(x)$ does not exist
- (b) $\lim_{x \rightarrow 0^-} f(x)$ does not exist
- (c) $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x)$
- (d) $\lim_{x \rightarrow 0} f(x) = 1$
- (e) $\lim_{x \rightarrow 0} |f(x)| = 1$

4. Let $f(x) = 2x + 1$. Which of the following statements is true?

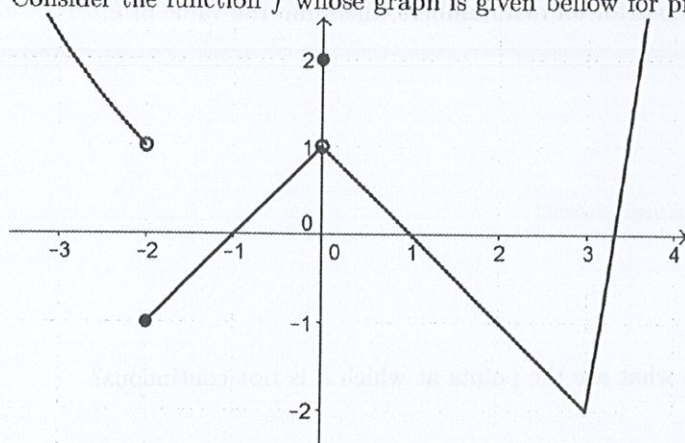
- (a) $\forall \epsilon > 0, \exists \delta > 0, \forall x \in \mathbb{R} \quad 0 < |x - 1| < \delta \implies |f(x) - 1| < \epsilon$
- (b) $\forall \epsilon > 0, \exists \delta > 0, \forall x \in \mathbb{R} \quad 0 < |x - 1| < \delta \implies |f(x) - 3| < \epsilon$
- (c) $\forall \epsilon > 0, \exists \delta > 0, \forall x \in \mathbb{R} \quad 0 < |x - 1| < \delta \implies |f(x) + 3| < \epsilon$
- (d) $\forall \epsilon > 0, \exists \delta > 0, \forall x \in \mathbb{R} \quad 0 < |x| < \delta \implies |f(x) - 3| < \epsilon$
- (e) None of the above

5. The set of upper bounds of the set $\{x : x < 1\}$ is

- (a) $(-\infty, -1)$
- (b) $(-\infty, -1]$
- (c) $(1, \infty)$
- (d) $[1, \infty)$
- (e) None of the above

Exam Code : PQ1ABC

Consider the function f whose graph is given below for problems 6-7.



6. Which of the following statements is/are true regarding the function f given above?

- (I) $\lim_{x \rightarrow 0} f(x) = 1$
- (II) $\lim_{x \rightarrow -2} f(x) = -1$.
- (III) $\lim_{x \rightarrow 1} f(x) = f(1)$

- (a) I only
- (b) II only
- (c) II and III only
- (d) I and III only
- (e) None of the above

7. Based on the graph of f , what is the set of points at which f is **not** continuous?

- (a) $\{-2\}$
- (b) $\{-2, 0\}$
- (c) $\{-2, 0, 3\}$
- (d) $\{0, 3\}$
- (e) None of the above

8. Let $g(x) = x^2 - 2x + 1$. Find $\lim_{h \rightarrow 0} \frac{g(2+h) - g(2)}{h}$.

- (a) $\frac{1}{h} (h^2 + 2h)$
- (b) 2
- (c) limit does not exist
- (d) $2x - 2$
- (e) None of the above

9. If $f(x) = \begin{cases} x^3 + k & \text{if } x \leq 1 \\ 2x^2 + 14 & \text{if } x > 1 \end{cases}$ is continuous for all real numbers, then find the value of k .

- (a) 3
- (b) 14
- (c) 15
- (d) 16
- (e) None of the above

10. If $h(x) = \begin{cases} x^3 + 3 & \text{if } x < 3 \\ x^2 + 18 & \text{if } 3 \leq x < 4 \\ x + 30 & \text{if } x \geq 4 \end{cases}$, then what are the points at which h is **not** continuous?

- (a) 3
- (b) 4
- (c) 3 and 4
- (d) All real numbers except 3 and 4
- (e) None of the above

02. Answer the following multiple choice questions on the answer sheet provided.

1. Find $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$.

- (a) 1
- (b) $-\infty$
- (c) ∞
- (d) Does not exist.
- (e) None of the above

2. Which of the following statements is/are true?

(I) $\frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy = 0$, where $F = F(x, y)$, is exact.

(II) $M(x, y)dx + N(x, y)dy = 0$ is exact if $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$.

(III) $M(x, y)dx + N(x, y)dy = 0$ is exact if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

- (a) I only
- (b) I and II only
- (c) II and III only
- (d) I and III only
- (e) None of the above

Exam Code : PQ1ABC

3. If $dF = 2xydx + x^2dy$ and $F(0,0) = 0$, then $F(x,y) =$

- (a) $2xy + x^2$
- (b) $2x^3y$
- (c) $x^2 + y$
- (d) x^2y
- (e) None of the above

4. What is the general solution to the second order linear equation $y'' + 9y = 0$? Here A and B are arbitrary constants.

- (a) Ae^{3x}
- (b) $A \cos(3x) + B \sin(3x)$
- (c) $Ae^{3x} + B$
- (d) $(Ax + B)e^{3x}$
- (e) None of the above

5. What is the differential equation of the family of curves given by the equation $\frac{1}{y} = Ax^2$, where A is an arbitrary constant?

- (a) $\frac{dy}{dx} = \frac{-2y}{x}$
- (b) $\frac{dy}{dx} = \frac{2x}{y}$
- (c) $\frac{dy}{dx} = yx^2$
- (d) $\frac{dy}{dx} = \frac{2y}{x}$
- (e) None of the above

6. What is an integrating factor for $y' - \frac{y}{x} = 4e^x$?

- (a) x
- (b) $-\frac{1}{x}$
- (c) $\frac{1}{x}$
- (d) e^x
- (e) None of the above

7. What is the solution to the initial value problem $3y' - \frac{y}{x} = x^3y^2$, $y(1) = 1$?

Hint: Use the substitution $v = y^{-1}$.

- (a) $y = -\frac{1}{13}x^4 + \frac{14}{13}x^{-\frac{1}{3}}$
- (b) $\frac{1}{y} = -\frac{1}{13}x^4 + \frac{14}{13}x^{-\frac{1}{3}}$
- (c) $y = -\frac{1}{11}x^4 + \frac{12}{11}x^{\frac{1}{3}}$
- (d) $\frac{1}{y} = -\frac{1}{11}x^4 + \frac{12}{11}x^{\frac{1}{3}}$
- (e) None of the above

8. What is the general solution to the ordinary differential equation $(e^x + 4y)dx + (4x - e^{-y})dy = 0$? Arbitrary constant in the solution is c .

- (a) $e^x + 4xy - ce^{-y} = 0$
- (b) $e^x + 4xy - e^{-y} = c$
- (c) $e^x + 4xy + e^{-y} = 0$
- (d) $e^x + 4xy + e^{-y} = c$
- (e) None of the above

9. Which of the following is an integrating factor of $(y^2 + xy)dx + (x^2 + 3xy)dy = 0$?

- (a) $\ln(y)$
- (b) $\ln(x)$
- (c) y
- (d) x
- (e) $\frac{1}{x}$

10. A population ($P = P(t)$) is decreasing over the time (t in seconds) such that at any given time, the rate change of the population is proportional to the square of the population at that time. When $t = 0$ the population is $P(0) = 1000$. It took 3 seconds to decrease the population to 500. What is the population after 4 seconds?

- (a) $\frac{3000}{7}$
- (b) $\frac{3000}{13}$
- (c) $\frac{300}{5}$
- (d) 250
- (e) None of the above

03. Write the answers, with all the steps, to this question on the writing paper that is provided.

1. If $a \in \mathbb{R}$ is such that $\forall n \in \mathbb{N}$, $|a| < \frac{1}{n}$, then prove that $a = 0$.
2. Find a $\delta > 0$ such that $\forall x \in \mathbb{R}$, $0 < |x - 2| < \delta \implies |x^2 - 4| < 0.01$.
3. Find the following limits.

(a) $\lim_{x \rightarrow \infty} (x + 1) \left(e^{(x+1)^{-1}} - 1 \right)$

(b) $\lim_{x \rightarrow 0^+} x \ln(x)$

(c) $\lim_{x \rightarrow \infty} \frac{e^x}{x^n}$, where $n \in \mathbb{N}$.

04. Write the answers, with all the steps, to this question on the writing paper that is provided.

1. Solve each of the following equations using the method of exact equations. If the given equation is not exact, first find an integrating factor.
 - (a) $(3x^2y + y^2x) dx + (x^3 + x^2y + y) dy = 0$
 - (b) $xdx + (x^2 + y^2 - 1) dy = 0$
2. Consider the equation $(y^3 - 3x^2y)dx + (2xy^2 - 2x^3)dy = 0$. Given that this equation has an integrating factor of the form $x^a y^b$, find a and b . Do not solve the equation completely.
3. Solve the linear differential equation $y'' + y' - 6y = x^2$, where $y = y(x)$.

———— @@@ ————



Problem 10: (10 points)

12. Write the answers, with all the steps, to this question on the writing paper that is provided.

1. If $n \in \mathbb{N}$ is such that $\forall x \in \mathbb{R}, f(x) < \frac{1}{n}$, then prove that $n = 0$.

2. Find $\delta > 0$ such that $\forall x \in \mathbb{R}, 0 < |x - 2| < \delta \implies |x^2 - 4| < 0.01$.

3. Find the following limits:

(a) $\lim_{x \rightarrow 1} (x^2 - 1) e^{\frac{1}{x-1}}$

(b) $\lim_{x \rightarrow 0} x \ln(x)$

(c) $\lim_{n \rightarrow \infty} \frac{1}{n^2}$, where $n \in \mathbb{N}$.

13. Write the answers, with all the steps, to this question on the writing paper that is provided.

1. Solve each of the following equations using the method of exact equations. If the given equation is not exact, first find an integrating factor.

(a) $(3x^2y + y^2x)dx + (x^3 + x^2y + y)dy = 0$

(b) $\sin(x) (x^2 + y^2 - 1)dy = 0$

2. Consider the equation $(y^2 - 2x)y' + (2xy^2 + 2x^2y) = 0$. Given that this equation has an integrating factor of the form $x^a y^b$, find a and b . Do not solve the equation completely.

3. Solve the linear differential equation $y' + y = x^2$, where $y = y(x)$.